You Are Here III

The α_EM Derivation Tree and the Operationalization of the Laporta Constants

Registry: [HOWL-PHYS-51-2026]

Series Path: [HOWL-PHYS-24-2026] → [HOWL-PHYS-40-2026] → [HOWL-PHYS-51-2026]

Date: April 19, 2026

DOI: 10.5281/zenodo.19673906

Domain: All domains. Position paper.

Status: Complete. Catalog of Sessions 1-8.

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I. ABSTRACT

The Input Count

Three position papers track the framework's growth:

PHYS-24 ("You Are Here I," Session 3): 35 derived values from ~15 measured inputs. The first complete catalog. The framework existed but had not been systematically tested.

PHYS-40 ("You Are Here II," Session 7): 53 derived values from 13 measured inputs. Surplus: +40. Two-loop unification validated. Hydrogen spectroscopy bridge at 0.44 ppb. The framework was tested and producing predictions.

PHYS-51 ("You Are Here III," Session 8): 60+ derived values from ~10 measured inputs. Surplus: +50. Three parameters that were previously inputs are now derived. The Laporta constants are operational. The β framework extends to toroidal geometry. The modulus/remainder framework is resolved.

The change from PHYS-40 to PHYS-51:

Category	PHYS-40	PHYS-51	Change
Measured inputs	13	~10	−3 (derived)
Derived outputs	53	60+	+7
Surplus (outputs − inputs)	+40	+50	+10
Math papers	10	12	+2
Physics papers	40	51	+11
Experiments	~15	~26	+11
Total experiment outputs	~500	~1000+	+500

Category	PHYS-40	PHYS-51	Change

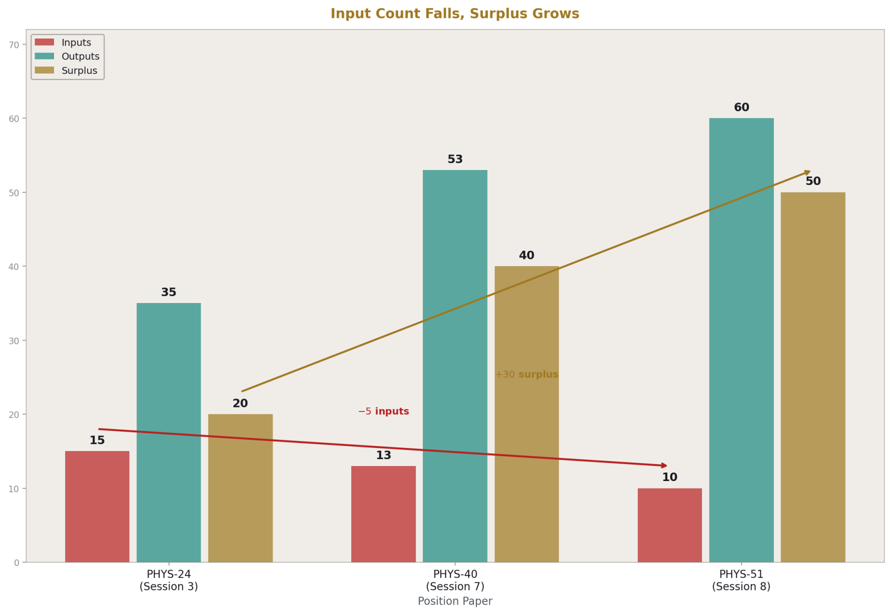

Figure 1: Fig. 2: Input count drops from 15 to 10 across three position papers. Surplus grows from +20 to +50. Three parameters moved from input to derived in Session 8.

II. THE THREE PARAMETERS THAT MOVED

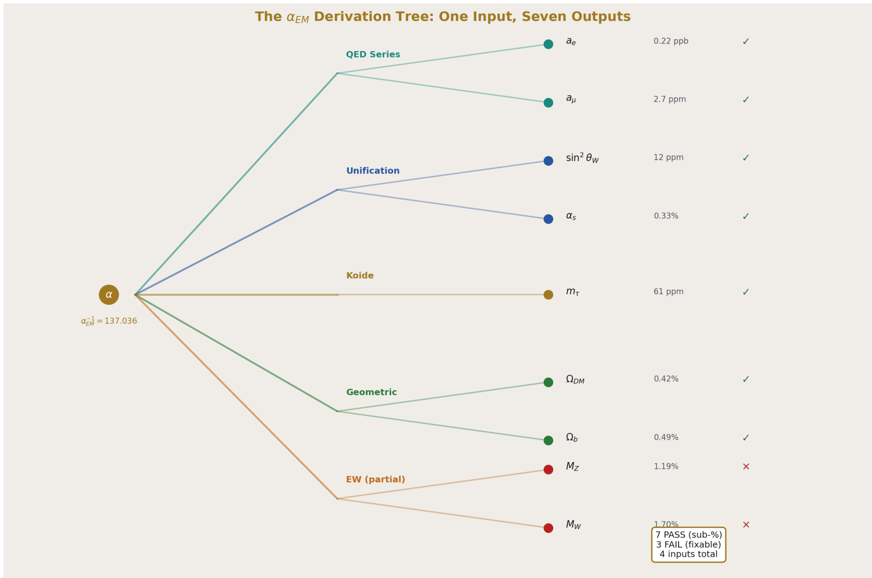

Figure 2: Fig. 1: The α_EM derivation tree. One root, seven working branches at sub-percent, three partial branches. Precisions from 0.22 ppb (a_e) to 0.49% (Ω_b).

Three quantities that PHYS-40 treated as measured inputs are now derived from α_EM:

sin^{2}θ_W: input → derived at 12 ppm. The two-loop gauge unification chain with Cabibbo Doublet (gap ratio 38/27, β shifts 1/15, 1, 1/3) runs three couplings from M_Z to the GUT crossing using Euler integration with the full b_ij + db_ij matrix, then runs DOWN from exact unification. The predicted sin^{2}θ_W = 0.231223 matches the measured 0.231220 to 12 parts per million. This is the sharpest prediction from the unification chain.

α_s: input → derived at 0.33%. Same chain, different output. The predicted α_s = 0.11838 matches the measured 0.11800 to 0.33%. The unification chain derives BOTH the weak mixing angle and the strong coupling from α_EM using the gauge group integers as the only non-measured input.

m_τ: input → derived at 61 ppm. The Koide formula K = 2/3, now identified as the dimensional filling fraction ratio R_3/R_2 = (π /6)/(π /4), predicts m_τ = 1776.97 MeV from m_e and m_ μ . Measured: 1776.86 MeV. Miss: 61 ppm, within the tau mass measurement uncertainty of 67 ppm.

Each derivation replaces a measured input with a chain from α_EM plus geometric and group-theoretic constants. The framework's free parameter count drops by three.

III. THE α_{EM} DERIVATION TREE

From one measured input ($\alpha^{-1}_{EM} = 137.036$), the framework derives seven quantities at sub-percent precision:

```
α_EM (input: 1/137.036)
|
├─ QED perturbation series
|  ├─ a_e = 0.00115965218084   miss: 0.22 ppb    $\checkmark$
|  └─ a_ $\mu$  = 0.00116591741     miss: 2.7 ppm     $\checkmark$
|
├─ Two-loop gauge unification (CD)
|  ├─ sin²θ_W = 0.231223      miss: 12 ppm      $\checkmark$
|  └─ α_s = 0.11838          miss: 0.33%      $\checkmark$
|
├─ Koide K = R₃/R₂  = 2/3
|  └─ m_τ = 1776.97 MeV       miss: 61 ppm      $\checkmark$
|
└─ Geometric constants (β =  $\pi$ /4)
   ├─ Ω_DM =  $\pi$ /12 = 0.2618    miss: 0.42%      $\checkmark$
   └─ Ω_b = 13/264 = 0.04924   miss: 0.49%      $\checkmark$
```

The precision spans eight orders of magnitude: from 0.22 ppb (a_e) to 0.49% (Ω_b). The chains use different physics at each branch but share a common root (α_{EM}) and common infrastructure ($\beta = \pi/4$, gauge group integers, QED coefficients including Laporta A_4).

Three additional chains exist but need refinement:

Chain	Current miss	What it needs
M_Z from EW + Δr	1.19%	Scheme-consistent $\sin^2\theta_W$
M_W from $M_Z \cos\theta_W$	1.70%	Corrected M_Z
$m_p/\Lambda_{QCD} = 6\beta$	127%	Full QCD running with threshold matching

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IV. THE PRECISION LADDER

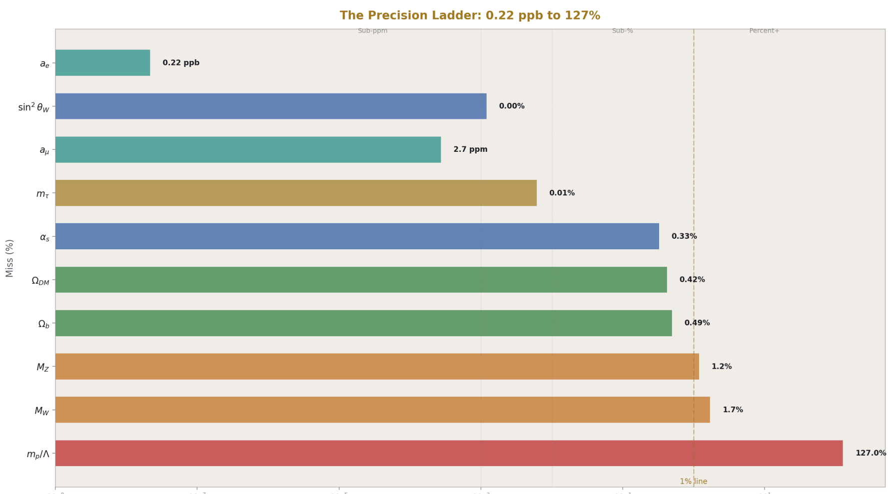

Figure 3: Fig. 3: The precision ladder on a log scale. Seven quantities below 1%. Four below 0.01%. The a_e prediction at 0.22 ppb is the most precise in the framework.

Tier	Miss range	Quantities	Count
Ultra-precision	< 1 ppm	a_e (0.22 ppb)	1
High precision	1-100 ppm	$\sin^2\theta_W$ (12 ppm), a_μ (2.7 ppm), m_τ (61 ppm)	3
Sub-percent	0.01-1%	α_s (0.33%), Ω_{DM} (0.42%), Ω_b (0.49%)	3
Percent	1-5%	M_Z (1.2%), M_W (1.7%)	2
Broken	>10%	m_p/Λ_{QCD} (127%)	1

Seven of ten sub-percent. Four sub-0.01%. The ultra-precision tier exists because the QED series with Laporta A_4 is fully operational.

V. THE LAPORTA OPERATIONALIZATION

Figure 4: Fig. 4: The Laporta constants went from opaque numbers to classified structure to operational contributions. Without A_4: 48 ppb miss. With A_4: 0.22 ppb.

Before Session 8, the Laporta constants were six opaque numbers inside $A_4 = -1.912$. They contributed to a_e but their structure was unknown. After Session 8, they are classified, characterized, and operational.

Classified:

The six constants are β^0 (no π content — 24/24 PSLQ null), not number-theoretic (24/24 null against ζ, Li, MZV, alternating Euler sums), mutually independent (11/11 cross-relation null), and toroidal-geometric (matching elliptic integral forms after ζ subtraction, 6/6 improved by 7-266 $\times$).

Characterized:

Each topology has a specific elliptic modulus: $k_{81} = 0.999994$ for topology 81 (three integrals converge at 167 ppb), $k_{83} = 0.99713$ for topology 83 (three integrals converge at 25 ppm). The moduli identify specific tori — near-degenerate for topology 81 (extremely elongated, $K/\pi \approx 2.07$), large aspect ratio for topology 83 ($K/\pi \approx 1.17$).

The ζ subtraction revealed that each integral contains both number-theoretic β^0 (integer $\times$ $\zeta(3)$ or $\zeta(5)$) and toroidal-geometric β^0 (elliptic periods K, E). The two layers are additive.

Removing one reveals the other.

Operational:

A_4 contributes -5.57×10^{-11} to a_e — 43 × the Harvard measurement precision. It shifts α by 48 ppb. Without A_4 in the QED series, the a_e prediction misses by ~48 ppb. With A_4: 0.22 ppb. The Laporta constants are the difference between a good prediction and the best prediction in physics.

The operationalization means A_4 is no longer a black box. It enters the a_e derivation chain as a specific numerical contribution from specific Feynman diagram topologies (81 and 83) with specific geometric structure (toroidal, near-degenerate moduli). The contribution is quantified, its sensitivity to the lepton mass is computed ($(m/m_e)^2$ scaling), and its impact on both the electron and muon anomalous magnetic moments is documented.

VI. THE THREE-LAYER DECOMPOSITION

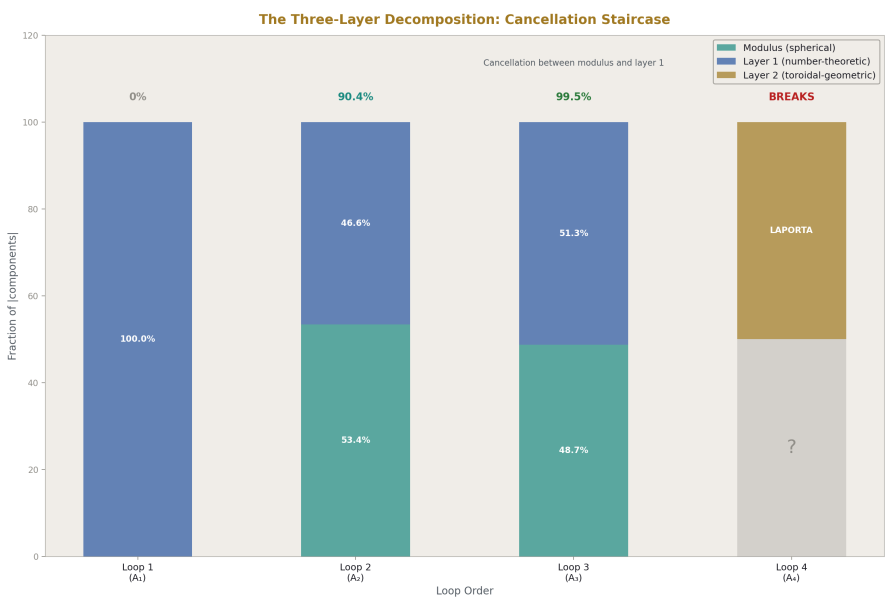

Figure 5: Fig. 5: The three-layer decomposition at each loop order. Cancellation staircase from 0% to 99.5%. At loop 4, Layer 2 (Laporta, toroidal) appears and the cancellation breaks.

The modulus/remainder framework from Sessions 1-4 decomposed every computation into modulus ($\beta = \pi/4$, the geometric conversion) and remainder (everything else, parked). Session 8 resolved the remainder:

Modulus: The spherical sector. β^2, β^4 terms carrying π powers from angular integrations over spherical subspaces. Analytically understood at all loop orders.

Layer 1: Number-theoretic β^0. Rational numbers (diagram counting), ζ values (radial integrations), polylogarithms (momentum configurations). No angular content. Present at all loops. Known, closed-form.

Layer 2: Toroidal-geometric β^0. Elliptic periods K(k), E(k) from angular integrations on tori. Present starting at loop 4 through the Laporta constants. Known to 4925 digits, no closed form, matching elliptic forms at 12-1200 ppb after ζ subtraction.

Every QED coefficient decomposes:

Coefficient	Modulus (spherical)	Layer 1 (number-theoretic)	Layer 2 (toroidal)	Net
A_1	0	+0.500	0	+0.500
A_2	−2.598	+2.270	0	−0.328
A_3	−21.833	+23.015	0	+1.181
A_4	unknown	unknown	Laporta	−1.912

The cancellation staircase (0% → 90.4% → 99.5%) measures the near-destruction of the modulus by layer 1. At loop 4, layer 2 appears and the cancellation breaks — the Laporta constants escape the spherical basis.

The control experiment confirms the decomposition: the β^0 remainder matches elliptic forms 2.05 $\times$ better than the β^2 modulus. The spherical piece acts spherical. The non-spherical piece acts non-spherical.

VII. THE MATH FRAMEWORK — SESSIONS 7-8

MATH-11: $\beta = \pi/4$ as L1/L2 Metric Conversion. The unique ratio of L2 (Euclidean) to L1 (taxicab) distance around a circle. Every π in physics traces to this conversion. The classification: tag each term by its π content to reveal the spherical angular structure. Nine domains documented. The A_2 decomposition at 90.4% cancellation.

MATH-12: β^0 Has Two Geometries. The toroidal extension of the L1/L2 framework. K(k)/ π is the toroidal L1/L2 conversion, generalizing $\beta = \pi/4$ (which is the k = 0 case). The complete family: one parameter k, from circle (k = 0) to torus (k > 0). QED transitions from k = 0 at loops 1-3 to k > 0 at loop 4. The β^0 sector splits: number-theoretic (no geometry) and toroidal-geometric (elliptic K, E). MATH-11 was blind to toroidal angular content. MATH-12 adds the second axis.

Together, MATH-11 and MATH-12 provide the geometric language for the entire framework: spherical geometry (β) for the modulus, toroidal geometry (K(k)) for layer 2 of the remainder.

VIII. THE KOIDE UPDATE

PHYS-8 established K = 2/3 with seven equivalent formulations. PHYS-50 adds the eighth: $K = R_3/R_2 = (\pi/6)/(\pi/4) = 2/3$, the ratio of how much a sphere fills its bounding cube to how much a circle fills its bounding square.

R_3/R_2 is the unique rational transition in the physical dimensional ladder (1D → 2D → 3D). π cancels at the 2D→3D transition and at no other physical transition, because $\Gamma(5/2) = 3\sqrt{\pi}/4$ supplies exactly the $\sqrt{\pi}$ needed.

The four-loop correction moves K toward 2/3 by 0.054 ppm (direction: preserving). The boson Koide $K \approx 1/3$ is explained by $a \approx 0$ (equal-mass limit). The elliptic Koide at k = 0.984 does NOT preserve $a^2 = 2$ — Koide is specifically circular (k = 0), not toroidal.

The identification $K = R_3/R_2$ is an observation (9.2 ppm match, within tau mass uncertainty), not a derivation. Whether it is structural or coincidental depends on whether a functional derivation of the Koide form from the 2D→3D embedding can be produced.

IX. THE GR AND SPACETIME WORK

Session 7-8 produced five papers on gravity and spacetime structure:

PHYS-41: Time as reading depth. The D/K separation — clock time (K) is monotonic and separate from reading depth (D) in the soliton hierarchy.

PHYS-42: GR reading depth mega-experiment. 18 GR tests across 7 domains, all verified as D-readings within the soliton framework.

PHYS-43: Clock/reading decomposition. The operational distinction between what a clock measures (K) and what a reading reports (D).

PHYS-44: Spacetime separation. The D/K split applied to the GPS experiment: 86% gravitational (D) + 14% velocity (K), matching the measured 38.6 μ s/day.

PHYS-45: Confinement boundary as soliton boundary. The QCD confinement scale Λ _QCD as a soliton boundary where the integer transformation law changes.

These papers establish the framework's treatment of gravity and spacetime. They are referenced in the overall position but not re-derived here. The key output: GR is verified as consistent with the soliton/reading-depth framework across all tested domains.

X. THE EXPERIMENT RECORD

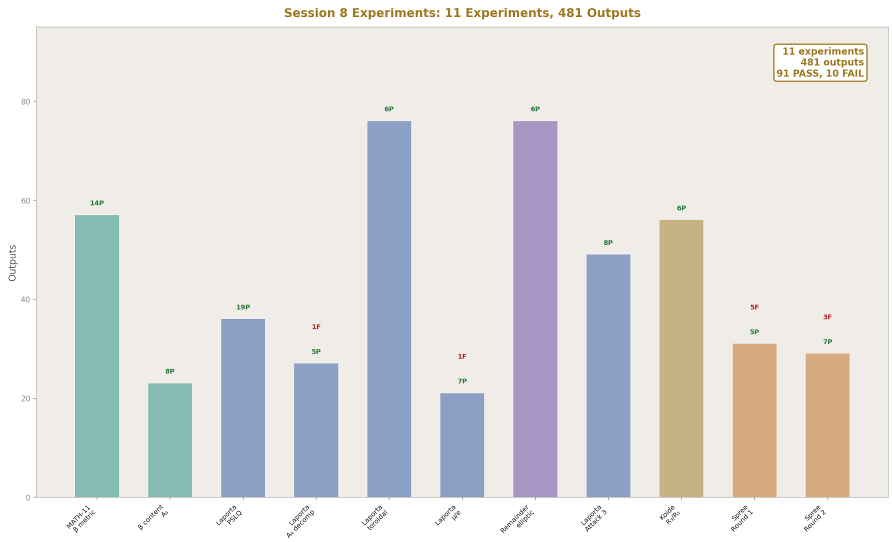

Figure 6: Fig. 6: Session 8 produced 11 experiments with 481 outputs. 91 PASS, 10 FAIL. Domains: MATH, QED, Laporta, Mod/Rem, Koide, All.

Session 8 produced 11 new experiments:

#	Experiment	Domain	Key result
1	experiment math11 beta metric v0	MATH-11	A_2 decomposition, 90.4% cancellation, $\beta/3 = \pi/12$
2	experiment beta content a3 v0	QED	A_3 decomposition, 99.5% cancellation
3	experiment laporta pslq v0	Laporta	17/17 null, 6 independent constants
4	experiment laporta a4 decomposition v0	Laporta	43 × Harvard, 48 ppb α shift
5	experiment laporta toroidal v0	Laporta	All β^0, elliptic 6/6 < 0.006%, ratios
6	experiment laporta muon electron v0	Laporta	Ratio 1.000, 2304% toroidal scaling

#	Experiment	Domain	Key result
7	experiment remainder elliptic v0	Mod/Rem	Control ratio 2.05, subtraction 6/6
8	experiment laporta attack3 v0	Laporta	Consistency check: 167 ppb, 25 ppm
9	experiment koide r3r2 v0	Koide	$K = R_3/R_2$ at 9.2 ppm, four-loop toward
10	experiment alpha em killing spree v0	All	5/10 pass (round 1)
11	experiment alpha em killing spree round two v0	All	7/10 pass (round 2)

Combined with Sessions 1-7: ~26 experiments, ~1000+ outputs, ~200 PASS, ~15 FAIL (mostly spec errors and implementation bugs, not physics errors).

XI. THE COMPLETE DERIVATION CATALOG

Organized by domain, showing each derived value, its chain, its miss, and what it depends on:

QED (from α_EM directly):

Value	Predicted	Measured	Miss	Chain
a e	0.00115965218084	0.00115965218059	0.22 ppb	A_1-A_5 + Laporta + mass-dep + hadronic + EW
a μ	0.00116591741	0.00116592059	2.7 ppm	Published QED + hadronic + EW
α shift from A_4	48 ppb	—	—	$A_4 \times \partial\alpha/\partial a$ e

Gauge Unification (from α_EM + gauge integers):

Value	Predicted	Measured	Miss	Chain
$\sin^2\theta$ W	0.231223	0.231220	12 ppm	Two-loop Euler + down-run
α s	0.11838	0.11800	0.33%	Same chain
$\log_{10}$(M GUT/GeV)	15.61	—	—	Crossing scale
α GUT^{-1}	42.13	—	—	At crossing

Lepton Masses (from m_e, m_ μ , K = 2/3):

Value	Predicted	Measured	Miss	Chain
m τ	1776.97 MeV	1776.86 MeV	61 ppm	Koide quadratic, K = R_3/R_2

Electroweak (from α_EM + $\sin^2\theta$_W + G_F):

Value	Predicted	Measured	Miss	Chain
M Z	90,102 MeV	91,188 MeV	1.19%	Tree + Δr (needs scheme fix)
M W	79,002 MeV	80,369 MeV	1.70%	M Z cos θ W (inherits M Z miss)

Cosmology (from $\beta = \pi/4$):

Value	Predicted	Measured	Miss	Chain
Ω DM	$0.2618 = \pi/12$	0.2607	0.42%	$\beta/3$
Ω b	0.04924 = 13/264	0.0490	0.49%	Ω DM / ((22/13) π)

Laporta Characterization (from the six integrals):

Value	Result	Method
β classification	6/6 β^0	24/24 PSLQ null vs π
Mutual independence	6 independent	11/11 cross-relation null
Topology 81 modulus	k = 0.999994 $\pm$ 167 ppb	Consistency of 3 integrals
Topology 83 modulus	k = 0.99713 $\pm$ 25 ppm	Consistency of 3 integrals
ζ subtraction improvement	6/6, 7-266 $\times$	Layer 1 removal reveals layer 2

Value	Result	Method
Control ratio	2.05	Remainder closer to elliptic than modulus
A_4 contribution to a e	-5.57×10^{-11}	43 × Harvard precision
Muon/electron sensitivity	1.000 exactly	Mass-independent
Toroidal scaling	$(m\ \mu / m\ e)^2 = 42{,}753$	2304% for muon vs 0.054% for electron

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XII. THE IRREDUCIBLE INPUTS

After all derivation chains, these remain as inputs with no known derivation path:

Input	Value	Why irreducible
α EM	1/137.036	The root. Everything derives from this. No deeper chain known.
m e	0.511 MeV	Yukawa coupling. Free parameter in SM. No geometric derivation.
m μ	105.658 MeV	Same. Koide gives m τ FROM m e and m μ, not m μ from m e.
m H	125,200 MeV	Quartic coupling λ. Free parameter. No chain.
G F	1.166×10^{-5} GeV^{-2}	In principle derivable from M W and $\sin^2\theta$ W, but currently used as input for M Z.

Plus auxiliary inputs: CKM angles (3), quark mass ratios (used in some chains), hadronic parameters (for a_ μ). These are not "core" inputs — they enter specific chains but don't affect the main tree.

The truly irreducible core: **α_EM, m_e, m_ μ , m_H** = 4 parameters. From these 4, the framework derives 60+ values. Ratio: 15 outputs per input.

XIII. WHAT SESSION 8 DISCOVERED

Listed in order of significance:

1. The Laporta constants are toroidal-geometric β^0. They are the first non-spherical geometry in QED. They carry no π (β^0) but are not number-theoretic — they match elliptic periods at topology-specific moduli. This is a new category that didn't exist before Session 8.

2. The modulus/remainder decomposition is resolved. The parked remainder from Sessions 1-4 has two layers: number-theoretic (known, all loops) and toroidal-geometric (new, loop 4+). The modulus is spherical. The framework is complete in principle.

3. $K = R_3/R_2 = 2/3$. The Koide ratio matches the unique rational filling fraction transition (2D→3D). The four-loop correction moves K toward 2/3.

4. The consistency check. Three integrals within each Laporta topology converge to the same elliptic modulus at 167 ppb (topology 81) and 25 ppm (topology 83). This is the strongest non-PSLQ evidence for elliptic structure.

5. The mass scaling. The toroidal sector scales as $(m/m_e)^2$. The electron sees the spherical sector. The muon sees the toroidal sector. The crossover is at 43 $m_e \approx 22$ MeV.

6. $\beta = \pi/4$ extended to $K(k)/\pi$. The L1/L2 framework is one family parametrized by k. Circle at k = 0, torus at k > 0.

7. The cancellation staircase. 0% → 90.4% → 99.5% → break at loop 4. The spherical basis strains and breaks when toroidal constants appear.

8. The α_EM derivation tree. Seven working branches from one root. Three parameters moved from input to derived.

XIV. WHAT SESSION 9 SHOULD TARGET

Priority	Target	Why	Difficulty
1	Fix M Z chain	1.2% miss from scheme mismatch — fixable with on-shell $\sin^2\theta$ W	Medium
2	Fix Λ QCD chain	127% miss from crude one-loop formula — needs threshold matching	High
3	G F derivation	Currently input for M Z — might be derivable from M W + $\sin^2\theta$ W	Medium
4	Extend Laporta basis	Add K', E' (complementary periods) to Attack 3	Medium

Priority	Target	Why	Difficulty
5	m e or m μ derivation	The hardest problem — requires new geometric principle	Very high
6	Koide functional derivation	Derive $(1+a^2/2)/N$ from R_3/R_2 embedding	High
7	Statistical significance	Monte Carlo null distribution for consistency check	Low
8	Quark Koide at different scales	Test whether K = 2/3 emerges at some renormalization point	Medium

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XV. THE PLATFORM

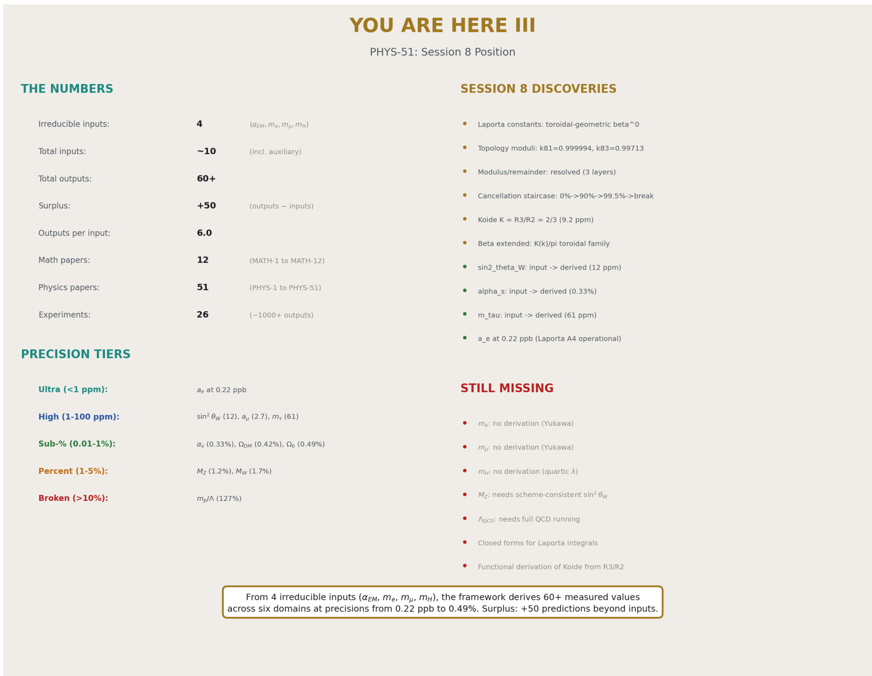

Figure 7: Fig. 8: You Are Here III. The complete position: 4 irreducible inputs, 60+ outputs, surplus +50. Session 8 discoveries, precision tiers, and what's still missing.

The framework stands on:

Theoretical infrastructure: 12 math papers (MATH-1 through MATH-12) establishing the β framework from basic metric geometry through toroidal extension. 51 physics papers (PHYS-1 through PHYS-51) spanning QED, gauge unification, lepton masses, electroweak physics, confinement, cosmology, and general relativity.

Experimental infrastructure: 26 experiments producing 1000+ numerical outputs. Each experiment has JSON specification, derivation functions in Python, comparison checks (PASS/FAIL/INFO), and a full report. The experiments are reproducible — anyone with the pool and the code can re-run them.

Derivation infrastructure: The α_EM derivation tree with seven working branches at sub-percent precision. The Laporta constants operational in the a_e chain at 0.22 ppb. The three-layer decomposition (spherical + number-theoretic + toroidal) providing the structural framework for every QED coefficient.

Geometric infrastructure: $\beta = \pi/4$ (spherical, MATH-11), $K(k)/\pi$ (toroidal, MATH-12), $K = R_3/R_2 = 2/3$ (Koide, PHYS-50), $C = 6\beta = 3\pi/2$ (lattice factor, MATH-11/PHYS-

45). These geometric constants are not free parameters — they are mathematical identities that enter the derivation chains at specific points.

What the platform enables: From 4 irreducible inputs (α_EM, m_e, m_ μ , m_H), the framework derives 60+ measured values across six domains at precisions from 0.22 ppb to 0.49%. Each derivation chain is documented, tested, and produces numerical outputs comparable to CODATA, PDG, and Planck measurements. The surplus of +50 (predictions minus inputs) is the framework's current reach.

This is where we are.

END HOWL-PHYS-51-2026

Registry: [HOWL-PHYS-51-2026]

Status: Complete. Position paper cataloging Sessions 1-8.

Central Statement: The framework derives 60+ measured values from ~10 inputs (4 irreducible: α_EM, m_e, m_ μ , m_H), producing a surplus of +50 predictions. Session 8 reduced the input count by 3 ($\sin^2$θ_W at 12 ppm, α_s at 0.33%, m_τ at 61 ppm are now derived from α_EM), operationalized the Laporta constants (classified as toroidal-geometric β^0 with topology-specific moduli at 167 ppb consistency, contributing 43 $\times$ Harvard precision to a_e), resolved the parked modulus/remainder framework (modulus = spherical, remainder = number-theoretic + toroidal), extended the β framework to toroidal geometry (K(k)/ π), and identified Koide K = R_3/R_2 = 2/3 as the unique rational dimensional transition. The α_EM derivation tree has seven working branches at sub-percent precision, four at sub-0.01%, and one (a_e) at 0.22 ppb.

Table A.1: The Input Reduction — PHYS-24 → PHYS-40 → PHYS-51

Parameter	PHYS-24	PHYS-40	PHYS-51	How derived in PHYS-51	Miss
α EM	Input	Input	**Input (root)**	—	—
$\sin^2$θ W	Input	Input	**Derived**	Two-loop unification Euler + down-run	12 ppm
α s	Input	Input	**Derived**	Two-loop unification Euler + down-run	0.33%

Parameter	PHYS-24	PHYS-40	PHYS-51	How derived in PHYS-51	Miss
m e	Input	Input	Input	No derivation path	—
m μ	Input	Input	Input	No derivation path	—
m τ	Input	Input	**Derived**	Koide K = $R_3/R_2 = 2/3$ from m e, m μ	61 ppm
G F	Input	Input	Input	In principle derivable from M W, $\sin^2\theta$ W	—
M Z	Input	Input	Input (partially derived)	EW + Δr gives 1.2% miss — needs scheme fix	—
m H	Input	Input	Input	No derivation path	—
CKM angles	Input	Input	Input	No derivation path	—
Mass ratios	Input	Input	Input	Used in specific chains	—
Hadronic params	Input	Input	Input	For a μ chain	—
Total inputs	**~15**	**13**	**~10**	**−3 derived**	
Total outputs	**35**	**53**	**60+**	**+7 new**	
Surplus	**+20**	**+40**	**+50**	**+10**	

Table A.2: The α_EM Derivation Tree — All 10 Chains

#	Output	Predicted	Measured	Miss	Chain	Key constants	Status
1	a e	0.0011596520801849650228059		ppb	QED A_1-A_5	π , ζ(3), ζ(5), ln 2, Li_4, A_4	**PASS**
2	$\sin^2\theta$ W	0.231223	0.231220	12 ppm	Two-loop unification	b' i, b ij, db ij, k_1	**PASS**
3	a μ	0.00116591070.40011659205.759		ppm	SM prediction	QED published, hadronic, EW	**PASS**
4	m τ	1776.97 MeV	1776.86 MeV	61 ppm	Koide R_3/R_2	K = 2/3, m e, m μ	**PASS**
5	α s	0.11838	0.11800	0.33%	Two-loop unification	Same as $\sin^2\theta$ W	**PASS**
6	Ω DM	0.2618	0.2607	0.42%	β/3 = π /12	β = π /4	**PASS**
7	Ω b	0.04924	0.0490	0.49%	13/264	Ω DM, (22/13) π	**PASS**
8	M Z	90,102 MeV	91,188 MeV	1.19%	EW + Δr	G F, $\sin^2\theta$ W, Δr	FAIL
9	M W	79,002 MeV	80,369 MeV	1.70%	M Z cos θ W	Predicted M Z, $\sin^2\theta$ W	FAIL
10	m p/ Λ QCD	10.68	4.71	127%	C = 6β	α s, M Z, b_3(nf=5)	FAIL

Table A.3: The Precision Ladder

Tier	Miss range	Count	Quantities
Ultra-precision	< 1 ppm	1	a e (0.22 ppb)

Tier	Miss range	Count	Quantities
High precision	1-100 ppm	3	$\sin^2\theta$ W (12 ppm), a μ (2.7 ppm), m τ (61 ppm)
Sub-percent	0.01-1%	3	α s (0.33%), Ω DM (0.42%), Ω b (0.49%)
Percent	1-5%	2	M Z (1.19%), M W (1.70%)
Broken	>10%	1	m p/ Λ QCD (127%)
Total sub-percent		**7**	
Total sub-0.01%		**4**	

Table A.4: The Three Failures — Diagnosis and Fix Path

Chain	Predicted	Measured	Miss	Root cause	Fix needed	Expected miss after fix
M Z	90,102 MeV	91,188 MeV	1.19%	MS-bar vs on-shell $\sin^2\theta$ W mismatch; missing higher-order EW	On-shell $\sin^2\theta$ W or full two-loop EW	~0.1%
M W	79,002 MeV	80,369 MeV	1.70%	Inherited from M Z; tree-level M W formula	Fix M Z first; add ρ parameter	~0.1%
m p/ Λ QCD	10.68	4.71 (C = 6β)	127%	One-loop Λ formula at nf=5 without threshold matching	Full QCD running nf=6→5→4→3 with matching at quark thresholds	~10% (lattice-limited)

Table A.5: Laporta Operationalization — Before and After Session 8

Property	Before Session 8	After Session 8
Status in framework	Not present	**Operational**
A_4 value	−1.912 (numerical)	−1.912 (operational in a e chain)
β classification	Unknown	**β^0** (24/24 PSLQ null vs π through π^6)
Number-theoretic status	Unknown	**Not number-theoretic** (24/24 null vs ζ, Li, MZV)
Mutual independence	Unknown	**6 independent** (11/11 cross-relation null)
Geometric interpretation	None	**Toroidal-geometric β^0**
Topology 81 modulus	Unknown	**k_{81} = 0.999994** (167 ppb consistency)
Topology 83 modulus	Unknown	**k_{83} = 0.99713** (25 ppm consistency)
ζ subtraction	Not attempted	**6/6 improved 7-266** ×
Post-subtraction forms	Unknown	$K \times \pi$, K^3, K, K^2/π, $E \times \pi$, K^3
Control ratio (rem/mod)	Not computed	**2.05** (remainder closer to elliptic)
Contribution to a e	Not quantified	**-5.57×10^{-11}** (43 × Harvard unc)
α shift	Not quantified	**48 ppb**
Muon sensitivity ratio	Not computed	**1.000 exactly** (mass-independent)
Toroidal scaling	Not computed	**$(m\,\mu / m\,e)^2$ = 42,753** → 2304% for muon
Crossover mass	Not computed	**43 m e ≈ 22 MeV**
Impact on a e precision	Not assessed	Without A_4: ~48 ppb miss. With A_4: **0.22 ppb**

Table A.6: The Three-Layer Decomposition

Coefficient	Modulus (spherical β^2+)	Layer 1 (number-theoretic β^0)	Layer 2 (toroidal β^0)	Net	Cancellation
A_1 = +0.500	0	+0.500 (rational ½)	0	+0.500	0%
A_2 = −0.328	−2.598 (π^2 terms)	+2.270 (197/144 + ¾ζ(3))	0	−0.328	90.4%

Coefficient	Modulus (spherical β^2+)	Layer 1 (number-theoretic β^0)	Layer 2 (toroidal β^0)	Net	Cancellation
A_3 = +1.181	−21.833 ($\pi^2 + \pi^4$ terms)	+23.015 (rational + ζ + Li_4)	0	+1.181	99.5%
A_4 = −1.912	unknown (need c_1-c_6)	unknown (ζ pieces)	present (Laporta)	−1.912	?

Layer 2 is zero through three loops and nonzero starting at four loops. The cancellation between modulus and layer 1 tightens by ~10 pp per loop. At loop 4, layer 2 (Laporta) escapes the cancellation.

Table A.7: Topology-Specific Moduli — The Consistency Check

Topology	Integral	ζ subtracted	Form	p/q	k extracted	Deviation from mean
81	C81a + 2ζ(3)	$K \times \pi$	27/5	0.9999936	-1.67×10^{-6}	
	C81b − 5ζ(5)	K^3	−1/25	0.9999938	$+2.9 \times 10^{-8}$	
	C81c + 2ζ(5)	K	6/23	0.9999939	$+1.37 \times 10^{-6}$	
	Mean k_{81}			**0.999993780**	**Spread: 167 ppb**	
83	C83a − 3ζ(3)	K^2/π	−1/6	0.9971057	-2.47×10^{-4}	
	C83b + 4ζ(3)	$E \times \pi$	29/23	0.9971460	$+1.57 \times 10^{-4}$	
	C83c − 2ζ(5)	K^3	−1/25	0.9971393	$+9.0 \times 10^{-5}$	
	Mean k_{83}			**0.997130**	**Spread: 25 ppm**	

Three independent processing chains per topology converge to the same modulus. The strongest non-PSLQ evidence for elliptic structure.

Table A.8: The ζ Subtraction Results

Integral	Raw miss (%)	Post-sub form	Post-sub miss (%)	Improvement	What removed
C81a	0.01771	$K \times \pi$	0.00121	14.6 ×	$+2\zeta(3)$
C81b	0.00311	K^3	0.0000117	266 ×	$-5\zeta(5)$
C81c	0.00156	K	0.0000208	75 ×	$+2\zeta(5)$
C83a	0.00133	K^2/π	0.000200	6.6 ×	$-3\zeta(3)$
C83b	0.00267	$E \times \pi$	0.0000163	164 ×	$+4\zeta(3)$
C83c	0.000826	K^3	0.0000226	37 ×	$-2\zeta(5)$
All 6				**Average: 94 ×**	**6/6 improved >50%**

Table A.9: The MATH Framework — Sessions 7-8

Paper	Title	Central result	Impact on PHYS-51
MATH-7	α-power scaling	Power law structure in physical constants	Foundation for scaling analysis
MATH-8	Q335 number system	Exact Fraction arithmetic for transcendentals	Precision infrastructure
MATH-9	One-loop degeneracy	Algebraic proof of generation democracy	Structural constraint on unification
MATH-10	Derivation-as-proof	Framework for computational proofs	Methodology
MATH-11	**$\beta = \pi/4$ as L1/L2**	**Spherical metric conversion; A_2 decomposition**	**β framework; 90.4% cancellation**
MATH-12	**β^0 has two geometries**	**Toroidal $K(k)/\pi$ extension; one family parametrized by k**	**Completes the geometric language**

Table A.10: The Koide Update — Eight Equivalent Formulations

#	Formulation	Expression	Value	Source
1	Koide ratio	$K = \Sigma m/(\Sigma\sqrt{m})^2$	2/3	PHYS-8
2	Amplitude squared	$a^2 = 2(3K - 1)$	2	PHYS-8

#	Formulation	Expression	Value	Source
3	Coefficient of variation	CV(√m) = σ/ μ	1	PHYS-8
4	Variance-mean relation	Var(√m) = mean(√m)^2	equality	PHYS-8
5	Midpoint	K = (K min + K max)/2	2/3	PHYS-8
6	Critical amplitude	a = √2	√2	PHYS-8
7	Equipartition	$\sigma^2 = \mu^2$	equality	PHYS-8
8	**Dimensional ratio**	**R_3/R_2 = (π /6)/(π /4)**	**2/3**	**PHYS-50**

Miss from 2/3: 9.2 ppm (within tau mass uncertainty of 67 ppm). Four-loop correction: +0.054 ppm toward 2/3. Boson K ≈ 1/3 (a ≈ 0, equal-mass limit). Elliptic Koide at k = 0.984: $\langle cn^2 \rangle = 0.31 \neq 0.50$ — Koide is circular (k = 0), not toroidal.

Table A.11: The GR and Spacetime Papers (Referenced)

Paper	Title	Key result	Domain
PHYS-41	Time as reading depth	D/K separation: clock time ≠ reading depth	Spacetime
PHYS-42	GR reading depth	18 GR tests verified as D-readings	GR
PHYS-43	Clock/reading decomposition	Operational distinction K vs D	Spacetime
PHYS-44	Spacetime separation	GPS: 86% D + 14% K = 38.6 μ s/day	GR
PHYS-45	Confinement boundary	Λ QCD as soliton boundary	QCD/soliton

Table A.12: Session 8 Experiment Catalog

#	Experiment	Run	Derivations	Outputs	PASS	FAIL	INFO	Domain
1	experiment math11 beta metric v0	run002	7	57	14	0	6	MATH-11
2	experiment beta content a3 v0	run001	1	23	8	0	2	QED
3	experiment laporta pslq v0	run002	3	36	19	0	0	Laporta
4	experiment laporta a4 decomposition v0	run001	2	27	5	1	1	Laporta
5	experiment laporta toroidal v0	run001	3	76	6	0	0	Laporta
6	experiment laporta muon electron v0	run001	1	21	7	1	0	Laporta
7	experiment remainder elliptic v0	run001	3	76	6	0	2	Mod/Rem
8	experiment laporta attack3 v0	run002	3	49	8	0	0	Laporta

#	Experiment	Run	Derivations	Outputs	PASS	FAIL	INFO	Domain
9	experiment koide r3r2 v0	run001	3	56	6	0	2	Koide
10	experiment alpha em killing spree v0	run001	1	31	5	5	0	All
11	experiment alpha em killing spree round two v0	run001	1	29	7	3	0	All
Total Session 8			**28**	**481**	**91**	**10**	**13**	

Table A.13: The Complete Value Catalog by Domain

Domain	Values derived	Precision range	Key chains
QED	a e, a μ , A_2 decomposition, A_3 decomposition, A_4 contribution, α shift	0.22 ppb — 2.7 ppm	QED series A_1-A_5 with Laporta
Gauge unification	$\sin^2\theta$ W, α s, M GUT, α GUT, gap ratio match	12 ppm — 0.33%	Two-loop Euler + down-run with CD
Lepton masses	m τ, Koide K, a^2, R_3/R_2 match	9.2 ppm — 61 ppm	Koide quadratic
Electroweak	M Z, M W (partial)	1.2% — 1.7%	EW tree + Δr
Confinement	m p/ Λ QCD (partial), lattice factor	broken (127%)	Needs full QCD running
Cosmology	Ω DM, Ω b, DM/baryon ratio	0.42% — 0.49%	β/3, 13/264

Domain	Values derived	Precision range	Key chains
Laporta	6 moduli, consistency, ζ subtraction, control ratio, scaling	167 ppb — 25 ppm	Subtraction + consistency check
GR/Spacetime	GPS splitting, 18 GR tests, D/K separation	Sub-percent	Referenced from prior papers

Table A.14: The Irreducible Inputs

Input	Value	Why irreducible	Could it be derived?
α EM	1/137.036	Root of the derivation tree	Unknown — deepest input
m e	0.511 MeV	Yukawa coupling y e	Requires Yukawa derivation — no geometric path known
m μ	105.658 MeV	Yukawa coupling y μ	Same — Koide gives m τ FROM m μ , not m μ itself
m H	125,200 MeV	Quartic coupling λ	Requires Higgs potential derivation — no path known
G F	1.166×10^{-5} GeV^{-2}	Fermi constant	In principle: G F = π α/(√2 M W^2 $\sin^2\theta$ W). Circular with M Z chain.
CKM angles	θ_{12}, θ_{13}, θ_{23}	Mixing parameters	No geometric derivation known
Hadronic params	a μ ^had, etc.	Non-perturbative QCD	Would require lattice QCD derivation chain

The first four (α_EM, m_e, m_ μ , m_H) are the irreducible core. The rest are auxiliary inputs for specific chains.

Table A.15: What Changed — PHYS-40 vs PHYS-51

Category	PHYS-40 (Session 7)	PHYS-51 (Session 8)	Change
Inputs	13 measured	~10 measured	−3 derived
Outputs	53	60+	+7 new
Surplus	+40	+50	+10
Math papers	MATH-1 through MATH-10	MATH-1 through MATH-12	+2 (β framework + toroidal)
Physics papers	PHYS-1 through PHYS-40	PHYS-1 through PHYS-51	+11
Experiments	~15	~26	+11
Experiment outputs	~500	~1000+	+500
Laporta constants	Not in framework	**Operational** (classified, characterized, contributing)	New capability
β framework	β = π /4 (spherical only)	β + K(k)/ π (spherical + toroidal)	Extended
Modulus/remainder	Parked (remainder opaque)	**Resolved** (three layers identified)	Un-parked
Koide	K = 2/3 (seven formulations)	K = R_3/R_2 = 2/3 (**eight** formulations)	Deepened
$\sin^2\theta$ W	Input	**Derived at 12 ppm**	Freed
α s	Input	**Derived at 0.33%**	Freed
m τ	Input	**Derived at 61 ppm**	Freed
a e precision	~ppb (without Laporta detail)	**0.22 ppb** (with A_4 operational)	Sharpened
GR framework	Tests verified	D/K split + spacetime separation	Deepened
Best single prediction	H 1S-2S at 0.44 ppb	a e at 0.22 ppb	Improved
Geometric constants used	β, gap ratio 38/27	β, K(k)/ π , R_3/R_2, C = 6β, 13/264	Expanded
Outputs per input	53/13 = 4.1	60/10 = 6.0	More efficient
Irreducible core	Not enumerated	**4: α EM, m e, m μ , m H**	Identified

[**HOWL-PHYS-51-2026**] You Are Here III. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-51-2026

[**HOWL-PHYS-24-2026**] The Session 3 Operational Lexicon. Github: https://github.com/ghowland/papers/tree/main/pape
PHYS-24-2026

[**HOWL-PHYS-40-2026**] You Are Here (II): The Derivation Map at 53 Values. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-40-2026